

Requirement 2a:

In many applications you'll need to convert a number from its decimal representation to its hexadecimal representation and vice versa.

1. You're required to write a function `convert` that takes two parameters as follows:
 - a. The first parameter is an integer that takes one of two values: either 1 (indicating hexadecimal to decimal conversion) or 2 (indicating decimal to hexadecimal conversion).
 - b. The second parameter is either a hexadecimal string (for the first case) or a decimal number (for the second case).
2. In the case of hexadecimal to decimal conversion, the function converts a hexadecimal representation of a number into its decimal equivalent (e.g., 1A -> 26) and prints the decimal result.
3. In the case of decimal to hexadecimal conversion, the function converts a decimal representation of a number into its hexadecimal equivalent (e.g., 26 -> 1A) and prints the hexadecimal result (uppercase, no 0x prefix).
4. In case one parameter only is specified for the function, then the function defaults and converts the input into its decimal equivalent (i.e., assumes hexadecimal input).
5. Note: You do not need to perform any checks concerning the validity of the input. All the test cases are either valid decimal integers or valid hexadecimal sequences.

```
$ convert 1 1A
26
$ convert 2 255
FF
$ convert 00ff
255
```

Requirement 2b:

A parentheses sequence is balanced if every opening parenthesis '(' is matched with a later closing parenthesis ')', and parentheses are properly nested.

1. You're required to write a function `checkParen` that prints 1 if the parentheses sequence is balanced and 0 otherwise.
2. The function takes only one parameter representing a parentheses sequence to be tested.
3. For example: `((()))` -> 1, `((()` -> 0, `)()` -> 0. The empty string is considered balanced.

Note: You do not need to perform any checks concerning the validity of the input. All the test cases contain only '(' and ')' characters.

```
$ checkParen '((()))'
1
$ checkParen '(()'
0
```

SUBMISSION:

- The shell file should be named: "req_2_firstName_secondName_ID.sh"
- Functions should be named exactly like stated in the requirements above
- Do not add additional input parameters or swap the order of the parameters

Hints:

- The source command (the first command in the two snippet examples) can be used to load any functions into the current terminal so you can use the function directly as shown.
- If you performed any changes in the script, you need to re-run the source command again.
- Don't forget to put `#!/bin/bash` at the beginning of your scripts
- Don't forget to make your script executable